

Cas d'étude:

Assesing the local aplicabilty of Remote sensing – based forest attribute models

Luigui Andrey Ramirez Parra

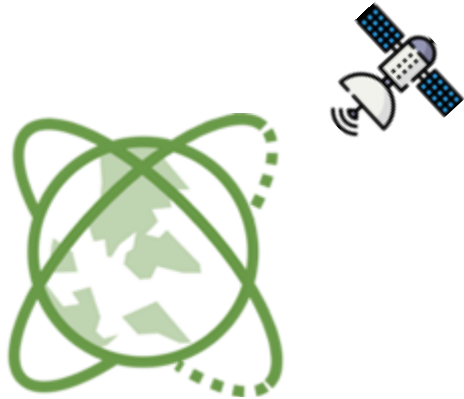
Supervisé par :

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Cedric Véga, Directeur de Recherche, Laboratoire d'inventaire forestier, ENSG, IGN

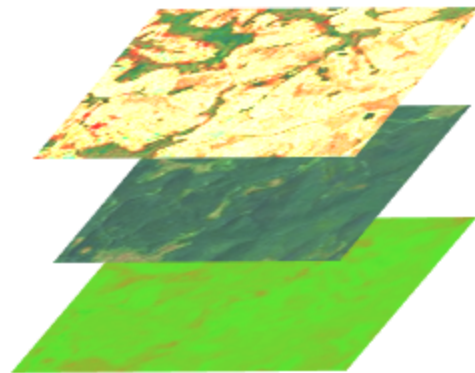


CONTEXTE

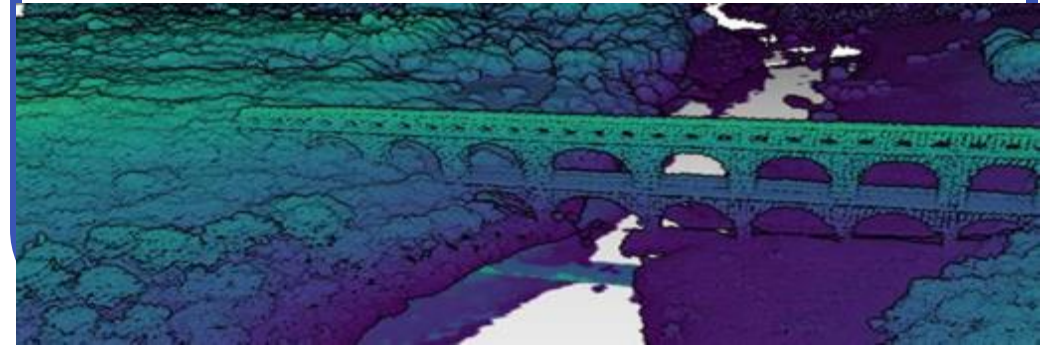
Les données de télédétection sont de plus en plus **nombreuses**



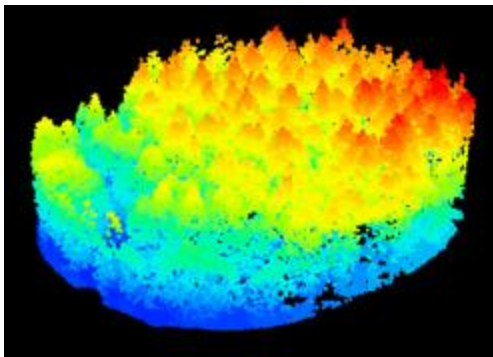
**Approche
Multi-source**



**LiDAR HD : vers une nouvelle
cartographie 3D du territoire**



Source: IGN



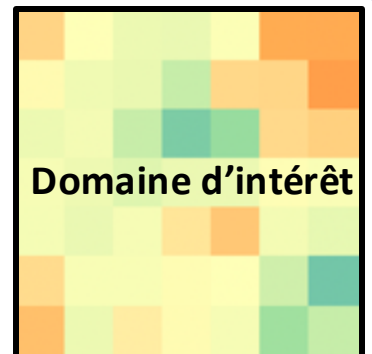
Avantages:

- ✓ Détail,
- ✓ Coûts,
- ✓ Temps
- ✓ Planification



Production de cartes:
Sur la base de modèles

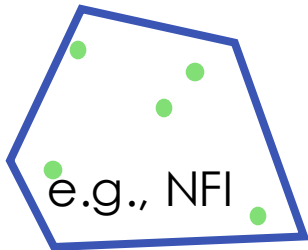
$$(Y) \sim F(X1) + \epsilon$$



CONTEXTE

Production de cartes:
Sur la base de modèles

Domaine de Calibration
clairsemé

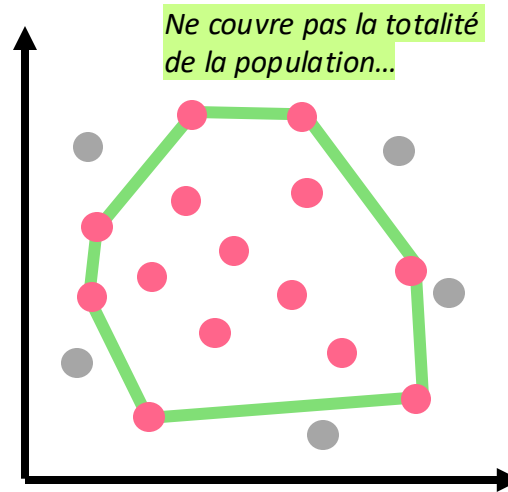


Biais local



Domaine de validité

L'approche du 'Convex Hull' (Brooks et al., 1988)



Prédictions
extrapolées

Affectent la
Transférabilité des
Modèles

Modèles **globalement**
non-biaisés

Préoccupation Majeure
Au niveau local

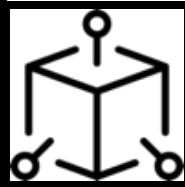
Prise de décision **gestion**
forestière

Sources: Vega et al., 2021; Bartley et al., 2019; Hsu et al., 2020; Poso et al., 1999 ; 3
James et al., 2013 ; Tomppo et al., 2008; Tomppo and Katila, 1991

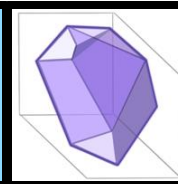
OBJECTIVES

Évaluer comment les modèles développés sur un grand territoire pour prédire la surface terrière produisent des biais locaux.

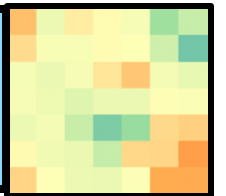
1). Évaluer la **transférabilité** des modèles pour prédire la Surface terrière.
(R2, RMSE, Biais)



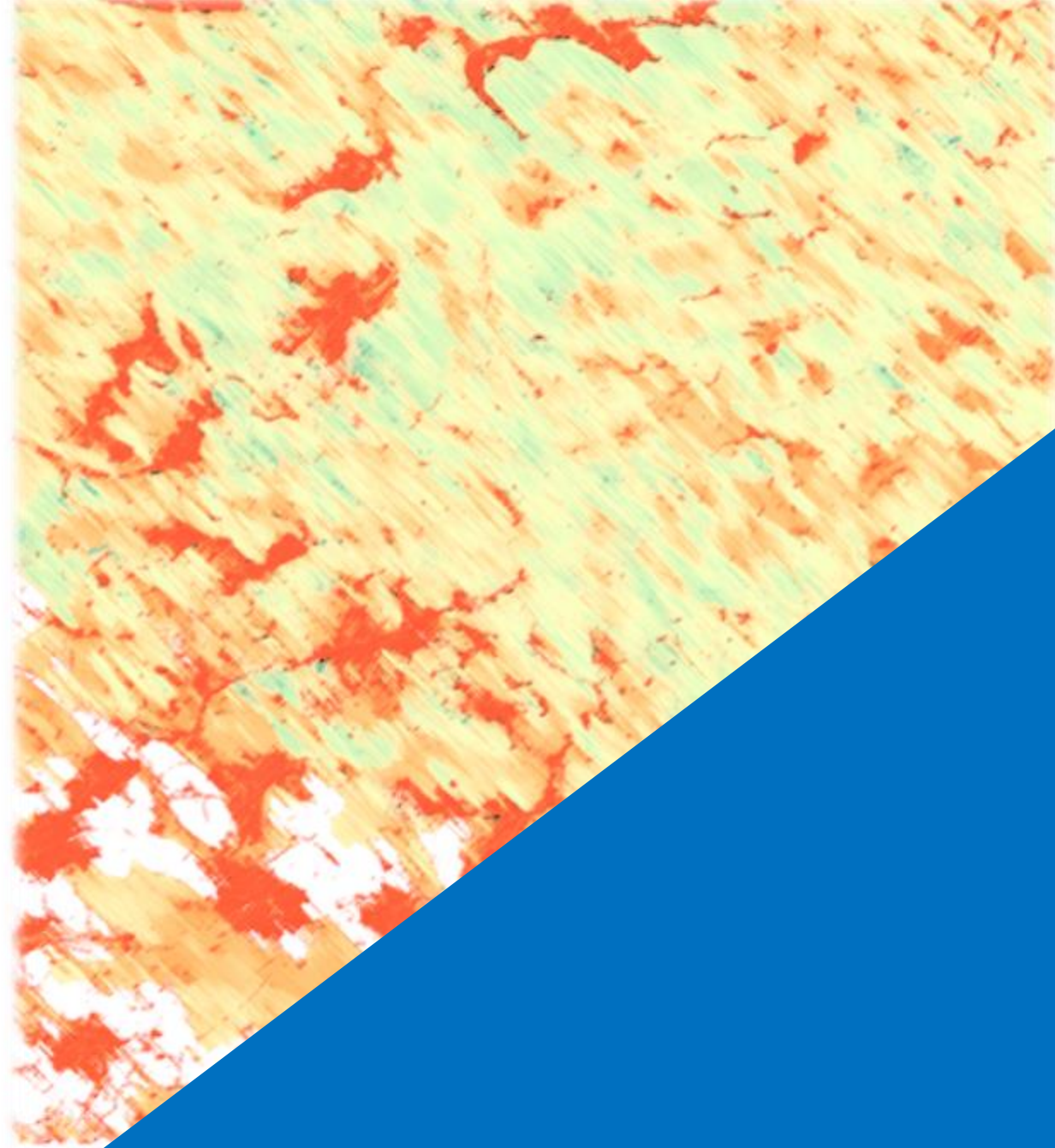
2). Tester comment la précision, le biais et la proportion de prédictions extrapolées se comportent en fonction de **L'intensité d'échantillonnage**.



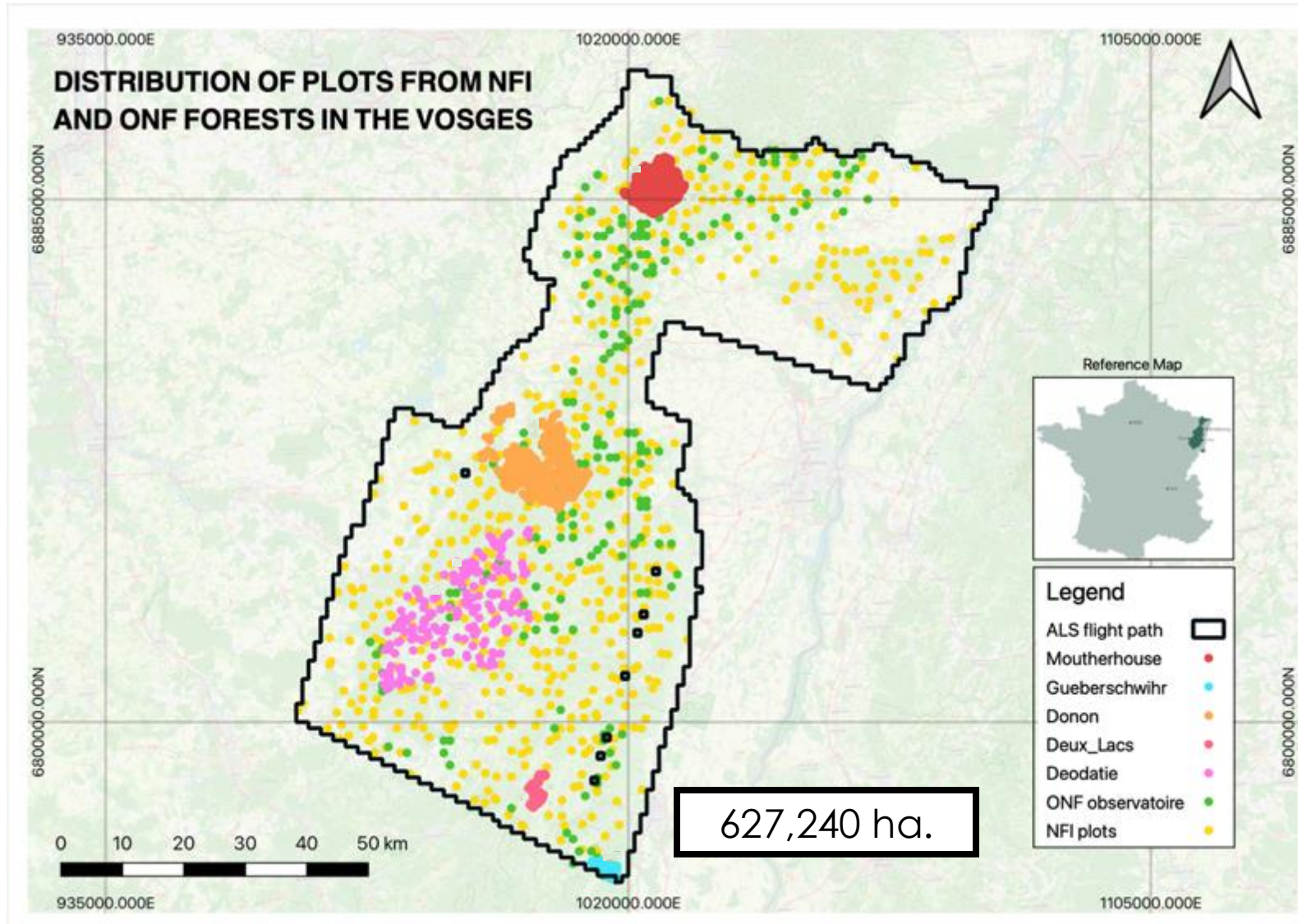
3). Classer, à l'aide du 'Convex hull', le **niveau d'extrapolation** sur les produits cartographiques



MATERIAUX ET MÉTHODES



ZONE D'ÉTUDE



Nombre de placettes

Forest	N of plots
Deodatie (2019)	192
Deux Lacs (2020)	346
Donon (2020-2021)	561
Guebenschwihr (2020-2021)	402
Mouthouse(2019)	544
Onf Observatory (2021)	198
NFI (2016-2020)	585
TOTAL	2824

L'Inventaire forestier national (IFN)

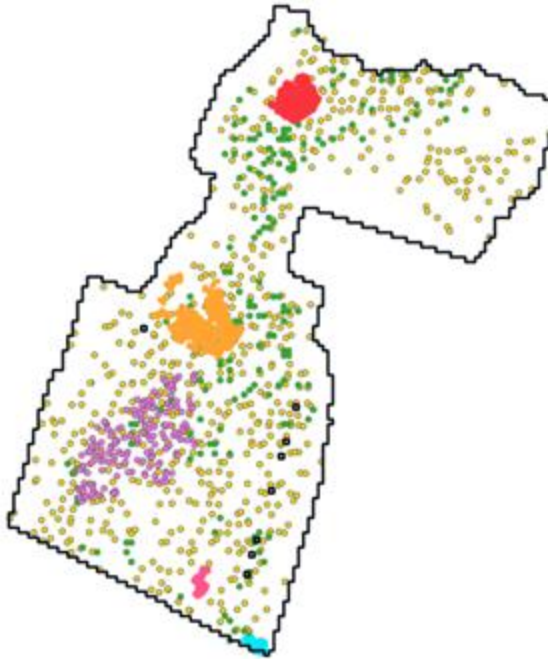


Arbres >17.5 cm diameter



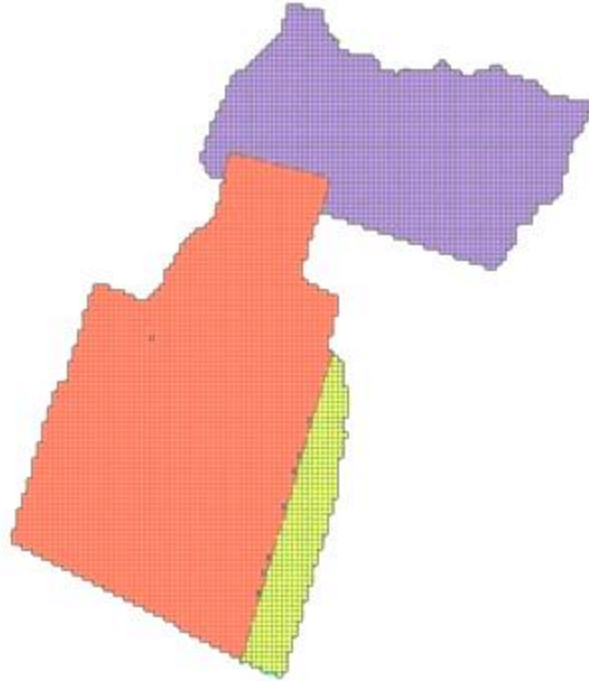
Données auxiliaires

Placettes
d'inventaire
forestier



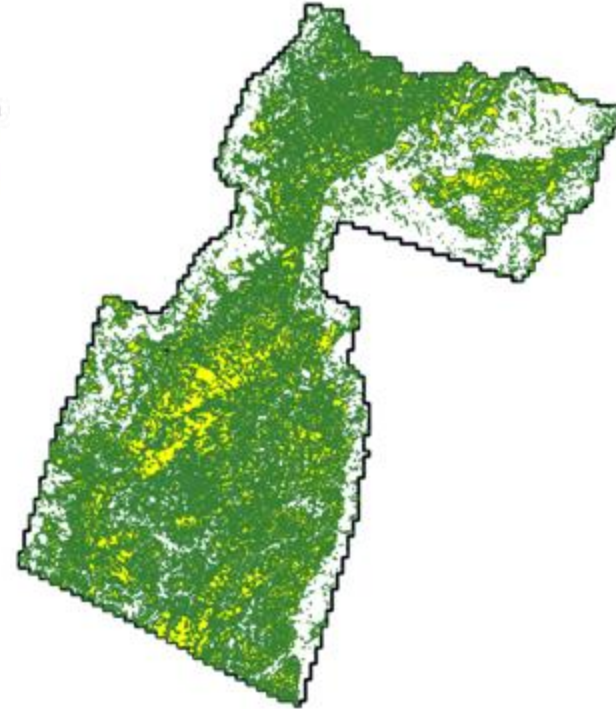
0 50km

3 Vols ALS
Métriques LIDAR
(2020-2021)



0 50km

Types de forêts
Bd Forêt 2.0



0 50km

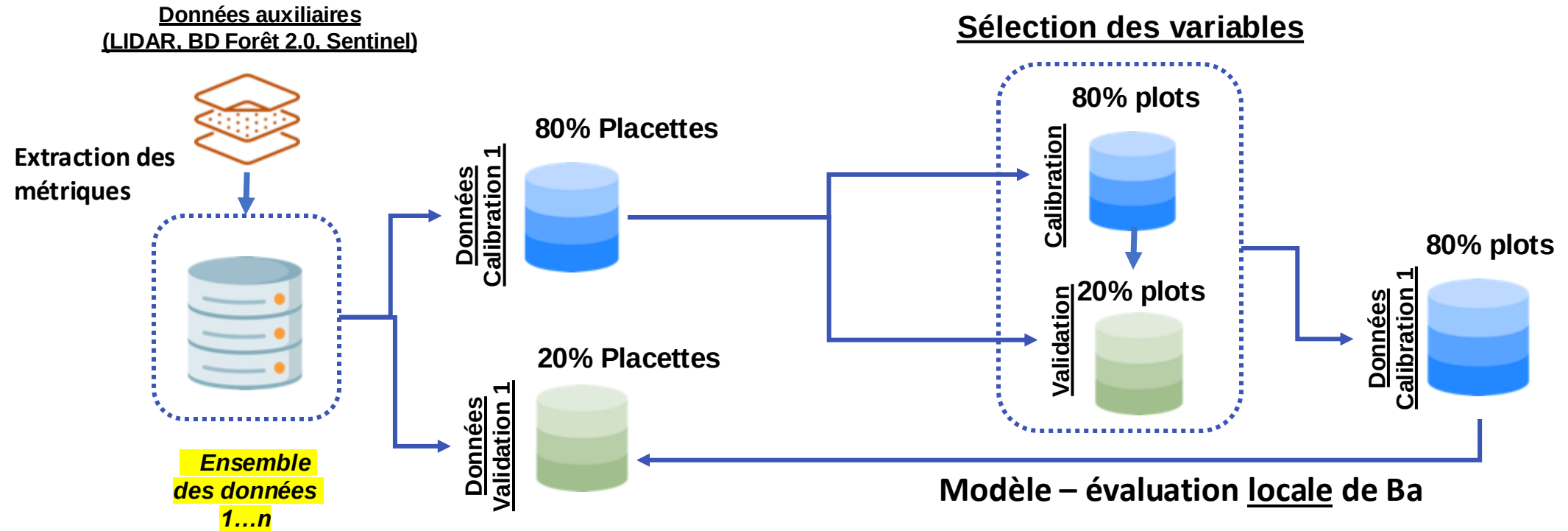
Image Sentinel
2019



0 50km

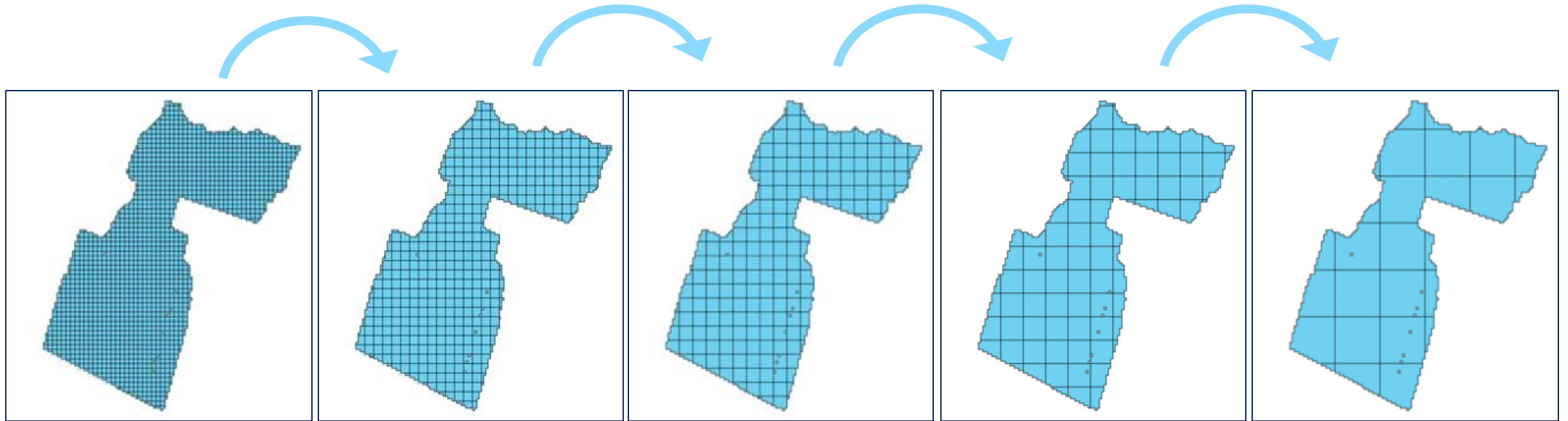
$$(\text{Ba}, \text{m}^2/\text{ha}) \sim \text{LiDAR}_1 + \text{Sentinel} + \text{Forest type}$$

ÉVALUATION DES MODÈLES



INTENSITE D'ECHANTILLONAGE

ONF Observatoire et IFN



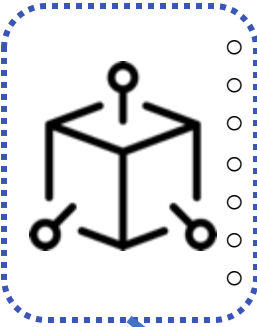
Résolution de la grille (km)	2	4	6	8	10	20
Nombre de Repetitions (bootstraps)	4	4	5	6	7	7

Qualité de l'ajustement

- ✓ RMSE
- ✓ BIAIS
- ✓ PIXELS EXTRAPOLÉS

PRODUCTION DE CARTES

Modèles



- Deodatie
- Deux Lacs
- Donon
- Gueberschwhir
- Mouterhouse
- ONF Observatoire
- NFI

CONVEX HULL



ZONE D'ÉTUDE



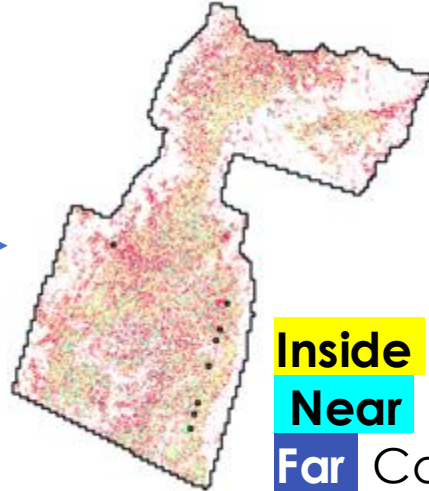
MODEL TRANSFERABILITY

Cartes: Ba (m²/ha) prediction



FIABILITÉ

Cartes: Niveau d'extrapolation

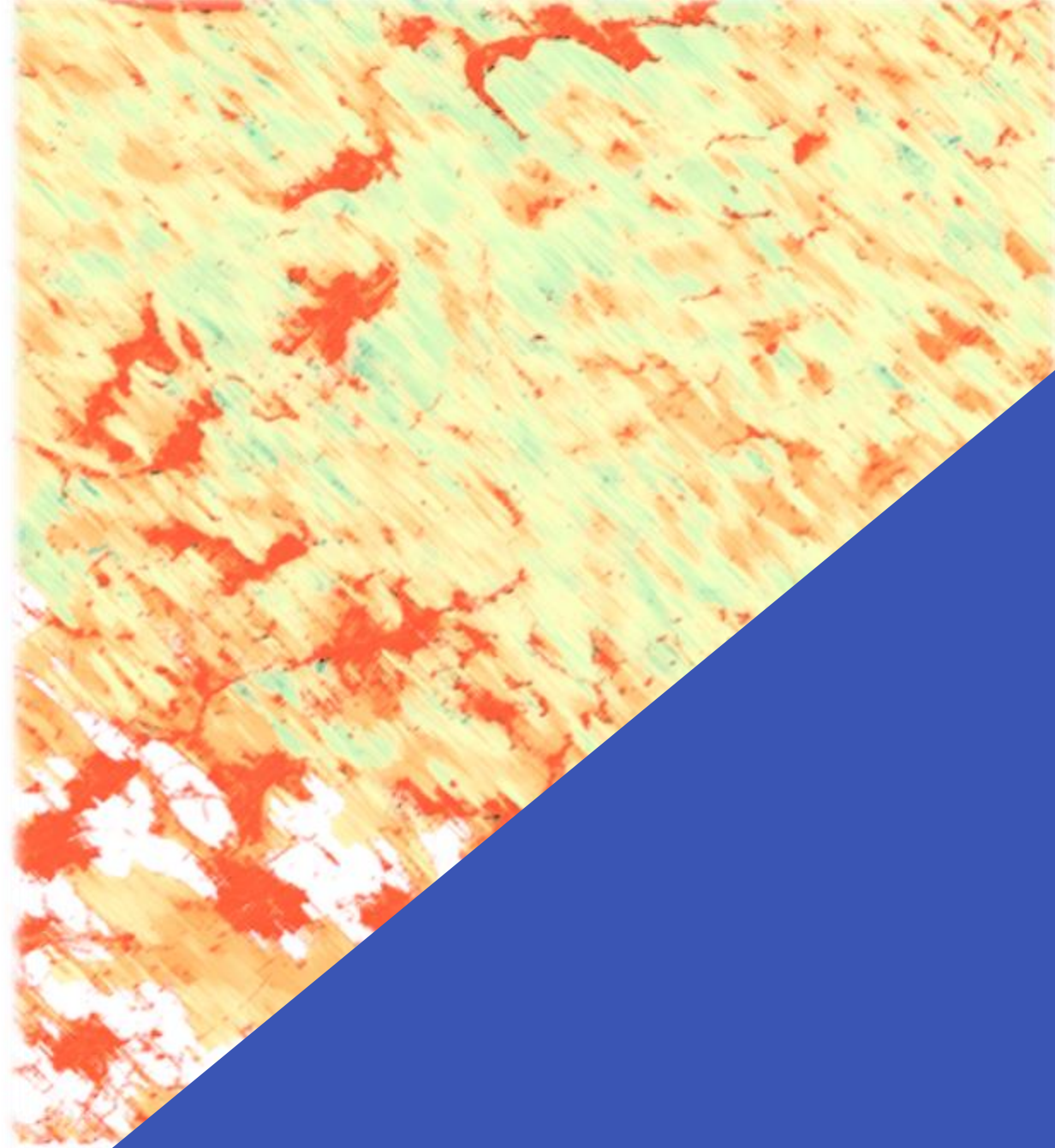


Distance Euclidienne (Données auxiliaires)

Distance Moyenne de Calibration

Inside Convex Hull
Near Convex Hull
Far Convex Hull

RESULTATS

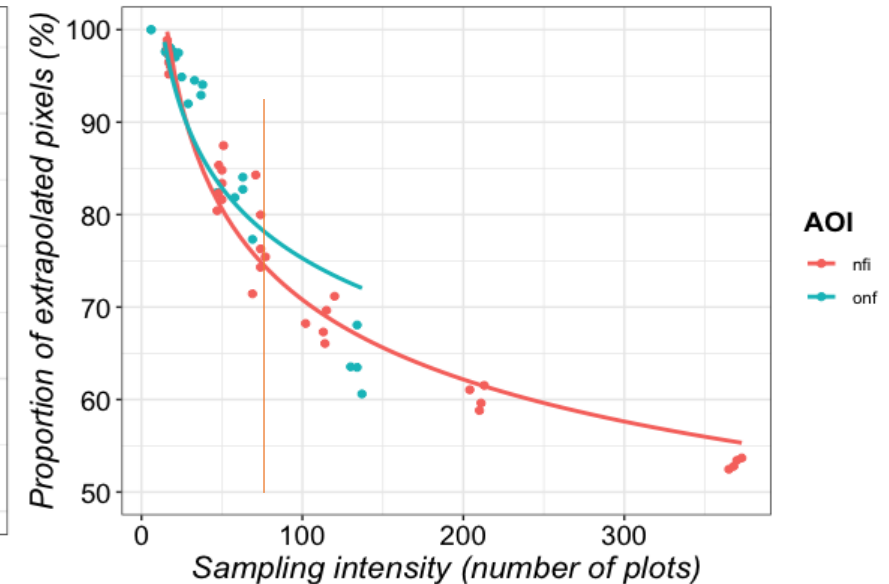
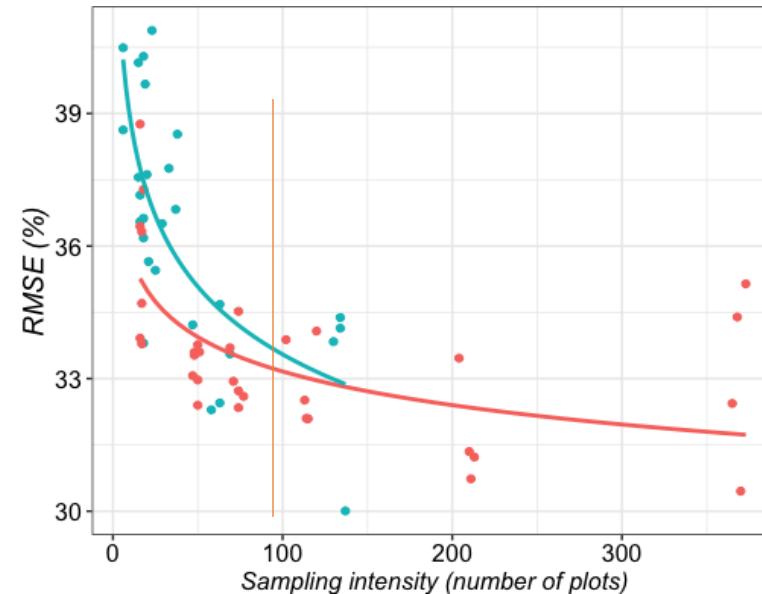
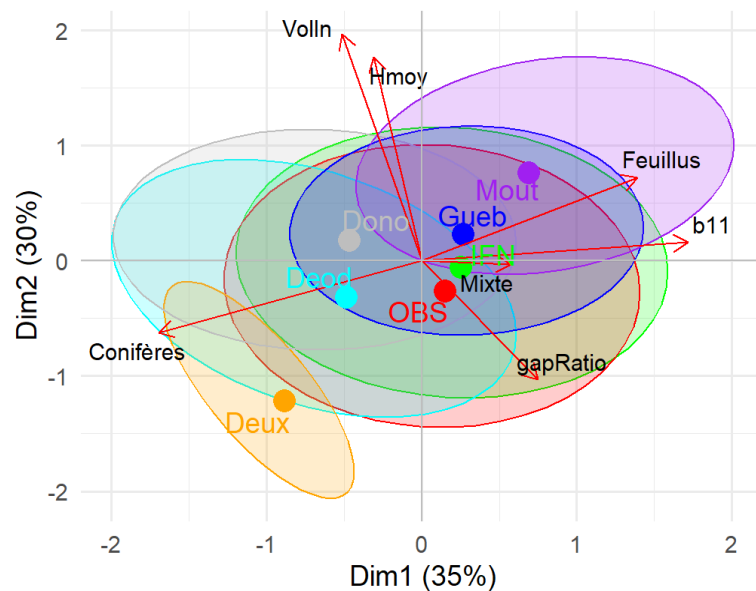


Transférabilité des Modelès

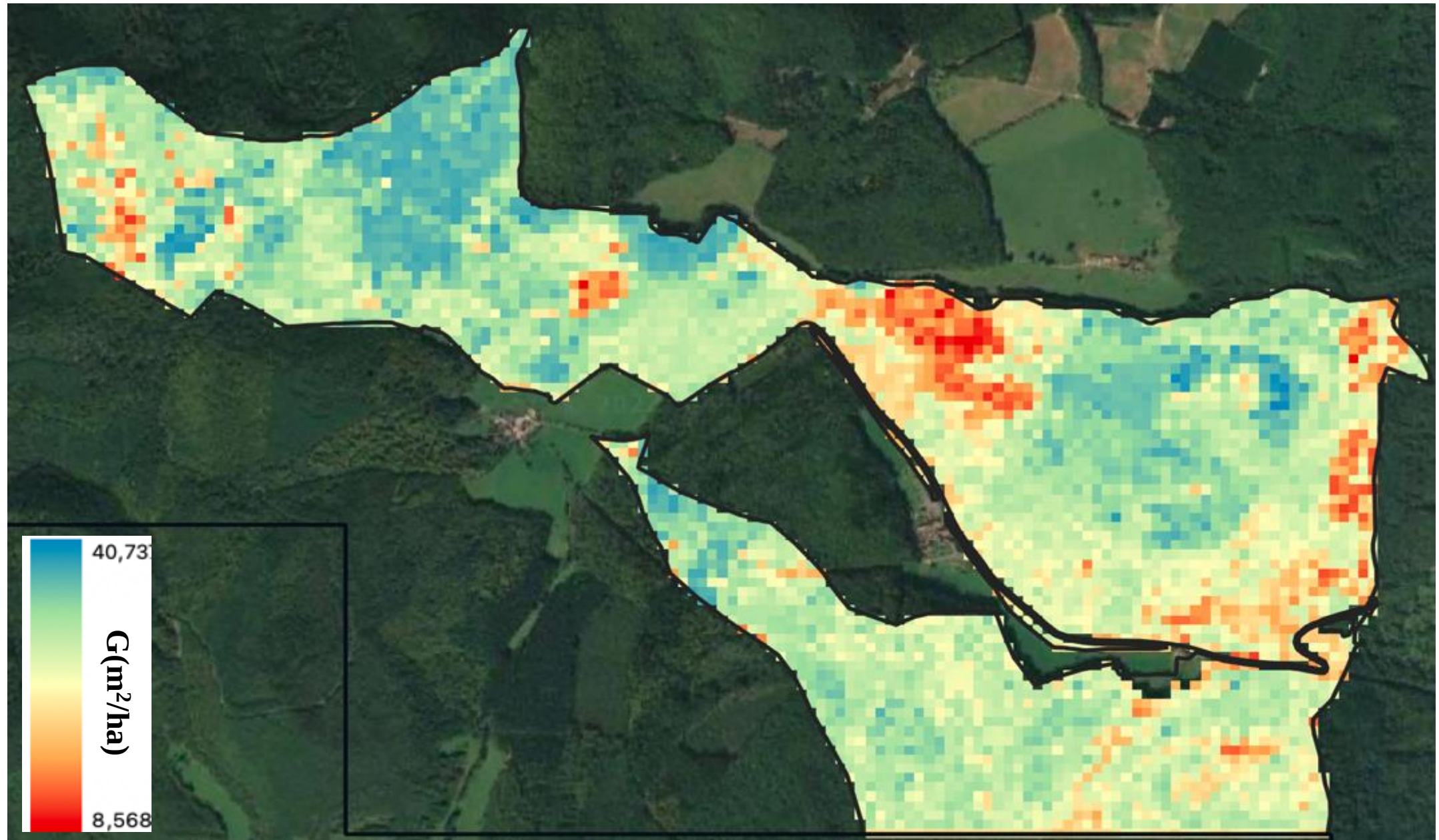
(R^2 , RMSE (m^2/ha) & BIAIS)

De les données de calibration (80% placettes), aux données de validation (20% placettes)

Transférabilité Forêt	Modèle											
	Observatoire ONF						IFN					
	R2	RMSE	RMSE (%)	Biais	Biais (%)	Extrap(%)	R2	RMSE	RMSE (%)	Biais	Biais (%)	Extrap(%)
Deodatie	0,53	9,5	35,0	-0,51	-1,9	28,1	0,59	14,1	51,8	-6,3	-23,0	31,8
Deux lacs	0,47	9,21	34,1	2,1	7,8	12,7	0,48	10,0	37,0	-3,5	-13,1	18,2
Donon	0,51	6,86	25,6	-2,16	-8,1	21,2	0,44	9,5	35,5	-7,3	-27,3	18,7
Gueberschwihr	0,32	7,1	25,2	0,57	2,0	15,2	0,23	7,8	27,6	-3,5	-12,3	16,2
Mouterhouse	0,30	6,92	29,3	-3,24	-13,7	26,1	0,22	9,8	41,6	-7,3	-31,1	20,0

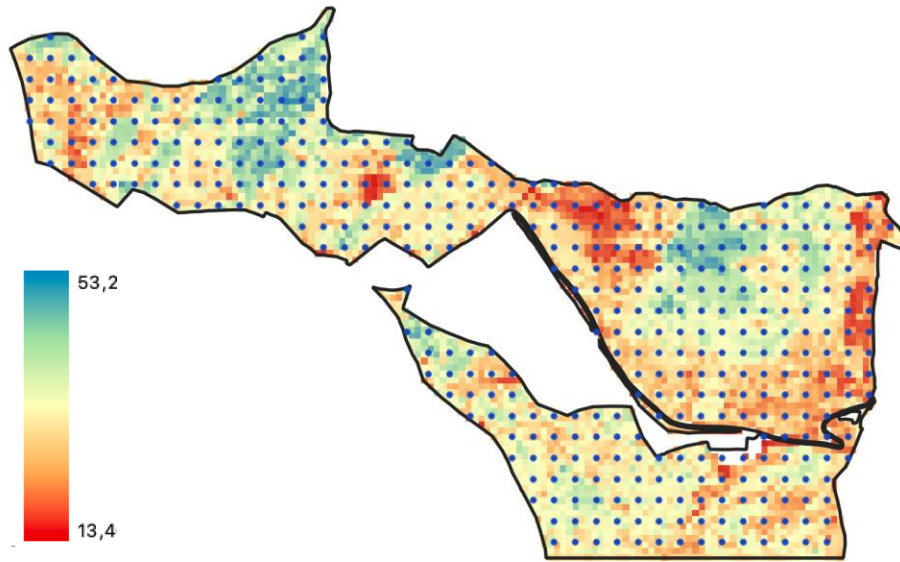


PRODUCTION DES CARTÉS (Gueberschwihr Forest)

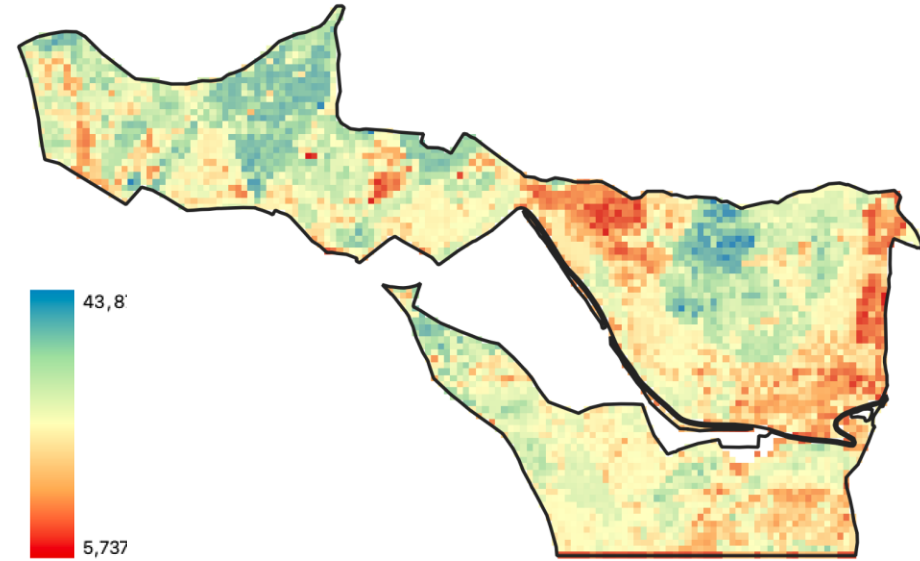


30m Pixel resolution

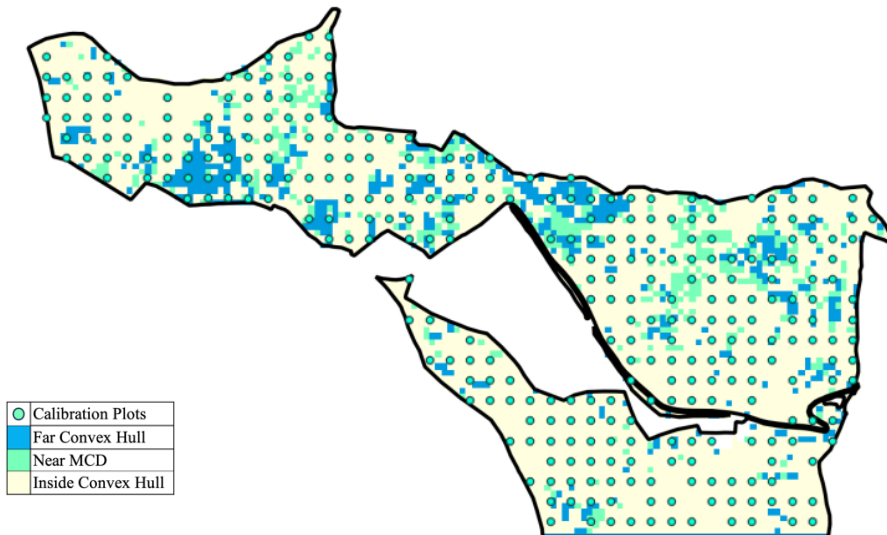
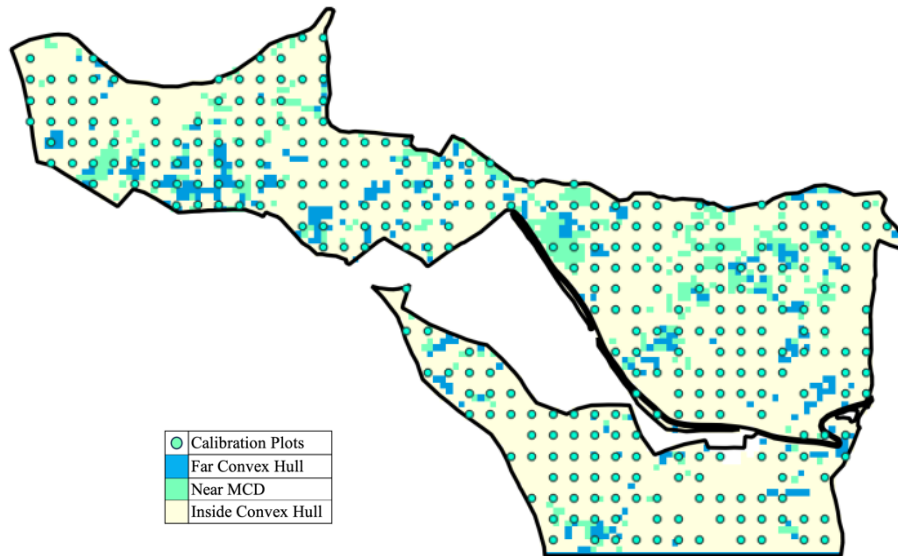
RELIABILITÉ DES CARTES (Forêt de GUEBERSCHWHIR)



Modelé IFN



Modelé ONF Observatoire



MESSAGES À RETENIR

- Les modèles du réseau régional de placettes appliqués dans les forêts locales perdent leur pouvoir prédictif à échelles locaux et vice versa
- Même avec un RMSE faible, le risque d'extrapolations élevées est présent à des intensités d'échantillonnage faibles
- Le 'Convex Hull' est une approche efficace qui pourrait compléter les décisions de gestion forestière, car elle localise les zones potentielles où les prédictions présentent un risque dans l'extrapolation.

L'avenir qui s'annonce...



Méthodes d'apprentissage profond / IA

- Modélisation des relations complexes entre LiDAR + Télédétection + Placettes inventaires , etc.
- Estimation pour petites zones (Small area estimation)
- Quantification des incertitudes
- Echantillonnage adaptatif .. Mieux représentation de variabilité local.

REFERENCES

- Babyak, M.A., 2004. What You See May Not Be What You Get: A Brief, Nontechnical Introduction to Overfitting in Regression-Type Models. *Psychosomatic Medicine* 66, 411–421.
- Bartley, M.L., Hanks, E.M., Schliep, E.M., Soranno, P.A., Wagner, T., 2019. Identifying and characterizing extrapolation in multivariate response data. *PLoS ONE* 14, e0225715. <https://doi.org/10.1371/journal.pone.0225715>
- Brooks, D.G., Carroll, S.S., Verdini, W.A., 1988. Characterizing the Domain of a Regression Model. *The American Statistician* 42, 187. <https://doi.org/10.2307/2684998>
- Coops, N.C., Tompalski, P., Goodbody, T.R.H., Queinnec, M., Luther, J.E., Bolton, D.K., White, J.C., Wulder, M.A., van Lier, O.R., Hermosilla, T., 2021. Modelling lidar-derived estimates of forest attributes over space and time: A review of approaches and future trends. *Remote Sensing of Environment* 260, 112477. <https://doi.org/10.1016/j.rse.2021.112477>
- Crookston, N.L., Finley, A.O., 2008. **yaImpute** : An R Package for k NN Imputation. *J. Stat. Soft.* 23. <https://doi.org/10.18637/jss.v023.i10>
- Fekety, P.A., Falkowski, M.J., Hudak, A.T., 2015. Temporal transferability of LiDAR-based imputation of forest inventory attributes. *Can. J. For. Res.* 45, 422–435. <https://doi.org/10.1139/cjfr-2014-0405>
- Fekety, P.A., Falkowski, M.J., Hudak, A.T., Jain, T.B., Evans, J.S., 2018. Transferability of Lidar-derived Basal Area and Stem Density Models within a Northern Idaho Ecoregion. *Canadian Journal of Remote Sensing* 44, 131–143. <https://doi.org/10.1080/07038992.2018.1461557>
- Frank, B., Mauro, F., Temesgen, H., 2020. Model-Based Estimation of Forest Inventory Attributes Using Lidar: A Comparison of the Area-Based and Semi-Individual Tree Crown Approaches. *Remote Sensing* 12, 2525. <https://doi.org/10.3390/rs12162525>

REFERENCES

- Gregoire, T.G., Ståhl, G., Næsset, E., Gobakken, T., Nelson, R., Holm, S., 2011. Model-assisted estimation of biomass in a LiDAR sample survey in Hedmark County, Norway This article is one of a selection of papers from Extending Forest Inventory and Monitoring over Space and Time. Can. J. For. Res. 41, 83–95. <https://doi.org/10.1139/X10-195>
- Guerra-Hernández, J., Botequim, B., Bujan, S., Jurado-Varela, A., Molina-Valero, J.A., Martínez-Calvo, A., Pérez-Cruzado, C., 2022. Interpreting the uncertainty of model-based and design-based estimation in downscaling estimates from NFI data: a case-study in Extremadura (Spain). GIScience & Remote Sensing 59, 686–704. <https://doi.org/10.1080/15481603.2022.2051383>
- Healey, S.P., Yang, Z., Cohen, W.B., Pierce, D.J., 2006. Application of two regression-based methods to estimate the effects of partial harvest on forest structure using Landsat data. Remote Sensing of Environment 101, 115–126. <https://doi.org/10.1016/j.rse.2005.12.006>
- Holmgren, P., Thuresson, T., 1998. Satellite remote sensing for forestry planning—A review. Scandinavian Journal of Forest Research 13, 90–110. <https://doi.org/10.1080/02827589809382966>
- IGN. 2022. A systematic nationwide inventory: on the whole French metropolitan territory. Institut National de L'information Géographique et Forestière-IGN. Accessed on June 2022. https://inventaire-forestier.ign.fr/IMG/pdf/180920_plaquette-methode_en.pdf
- James, G., Witten, D., Hastie, T., Tibshirani, R., 2013. An Introduction to Statistical Learning, Springer Texts in Statistics. Springer New York, New York, NY. <https://doi.org/10.1007/978-1-4614-7138-7>
- Kangas, A., Maltamo, M. (Eds.), 2006. Forest Inventory: Methodology and Applications, Managing Forest Ecosystems. Springer Netherlands, Dordrecht. <https://doi.org/10.1007/1-4020-4381-3>
- Poso, S., Wang, G., Tuominen, S., 1999. Weighting alternative estimates when using multi-source auxiliary data for forest inventory. Silva Fenn. 33. <https://doi.org/10.14214/sf.669>

REFERENCES

- Renaud, J.-P., Sagar, A., Barbillon, P., Bouriaud, O., Deleuze, C., Véga, C., 2022. Characterizing the calibration domain of remote sensing models using convex hulls. *International Journal of Applied Earth Observation and Geoinformation* 21.
- Renaud, J.-P., Vega, C., Durrieu, S., Lisein, J., Magnussen, S., Lejeune, P., Fournier, M., 2017. Stand-level wind damage can be assessed using diachronic photogrammetric canopy height models. *Annals of Forest Science* 74, 74.
<https://doi.org/10.1007/s13595-017-0669-3>
- Roussel, J.-R., Auty, D., Coops, N.C., Tompalski, P., Goodbody, T.R.H., Meador, A.S., Bourdon, J.-F., de Boissieu, F., Achim, A., 2020. lidR: An R package for analysis of Airborne Laser Scanning (ALS) data. *Remote Sensing of Environment* 251, 112061. <https://doi.org/10.1016/j.rse.2020.112061>
- Roussel, J.R., Barber, C.B., Habel, K., Grasman, R., Mozharovski, R.B., Steratt, P., 2022. geometry: Mesh Generation and Surface Tessellation R package version 0.4.5.
- Tokola, T., Heikkilä, J., 1997. Improving satellite image based forest inventory by using a priori site quality information. *Silva Fenn.* 31. <https://doi.org/10.14214/sf.a8511>
- Tokola, T., Pitkänen, J., Partinen, S., Muinonen, E., 1996. Point accuracy of a non-parametric method in estimation of forest characteristics with different satellite materials. *International Journal of Remote Sensing* 17, 2333–2351.
<https://doi.org/10.1080/01431169608948776>
- Tompalski, P., White, J.C., Coops, N.C., Wulder, M.A., 2019. Demonstrating the transferability of forest inventory attribute models derived using airborne laser scanning data. *Remote Sensing of Environment* 227, 110–124.
<https://doi.org/10.1016/j.rse.2019.04.006>

REFERENCES

- Tomppo, E., Goulding, C., Katila, M., 1999. Adapting Finnish Multi-Source Forest Inventory Techniques to the New Zealand Preharvest Inventory. *Scandinavian Journal of Forest Research* 14, 182–192. <https://doi.org/10.1080/02827589950152917>
- Tomppo, E., Katila, M., 1991. Satellite image-based national forest inventory of finland for publication in the igarss'91 digest, in: [Proceedings] IGARSS'91 Remote Sensing: Global Monitoring for Earth Management. Presented at the [IGARSS'91 Remote Sensing: Global Monitoring for Earth Management, IEEE, Espoo, Finland, pp. 1141–1144. <https://doi.org/10.1109/IGARSS.1991.579272>
- van Ewijk, K., Tompalski, P., Treitz, P., Coops, N.C., Woods (ret.), M., Pitt (ret.), D., 2020. Transferability of ALS-Derived Forest Resource Inventory Attributes Between an Eastern and Western Canadian Boreal Forest Mixedwood Site. *Canadian Journal of Remote Sensing* 46, 214–236. <https://doi.org/10.1080/07038992.2020.1769470>
- Vega, C., St-Onge, B., 2009. Mapping stand site index and age combining times-series of canopy height models with growth model. *Forest Ecology and Management* 257, 951–959.
- Véga, C., Renaud, J.-P., Durrieu, S., Bouvier, M., 2016. On the interest of penetration depth, canopy area and volume metrics to improve Lidar-based models of forest parameters. *Remote Sensing of Environment* 175, 32–42. <https://doi.org/10.1016/j.rse.2015.12.039>
- Vega, C., Renaud, J.-P., Sagar, A., Bouriaud, O., 2021. A new small area estimation algorithm to balance between statistical precision and scale. *International Journal of Applied Earth Observation and Geoinformation* 97, 102303. <https://doi.org/10.1016/j.jag.2021.102303>
- White, J.C., Coops, N.C., Wulder, M.A., Vastaranta, M., Hilker, T., Tompalski, P., 2016. Remote Sensing Technologies for Enhancing Forest Inventories: A Review. *Canadian Journal of Remote Sensing* 42, 619–641. <https://doi.org/10.1080/07038992.2016.1207484>
- Yates, K.L., Bouchet, P.J., Caley, M.J., Mengersen, K., Randin, C.F., Parnell, S., Fielding, A.H., Bamford, A.J., Ban, S., Barbosa, A.M., Dormann, C.F., Elith, J., Embling, C.B., Ervin, G.N., Fisher, R., Gould, S., Graf, R.F., Gregr, E.J., Halpin, P.N., Heikkinen, R.K., Heinänen, S., Jones, A.R., Krishnakumar, P.K., Lauria, V., Lozano-Montes, H., Mannocci, L., Mellin, C., Mesgaran, M.B., Moreno-Amat, E., Mormede, S., Novaczek, E., Oppel, S., Ortuño Crespo, G., Peterson, A.T., Rapacciuolo, G., Roberts, J.J., Ross, R.E., Scales, K.L., Schoeman, D., Snelgrove, P., Sundblad, G., Thuiller, W., Torres, L.G., Verbruggen, H., Wang, L., Wenger, S., Whittingham, M.J., Zharikov, Y., Zurell, D., Sequeira, A.M.M., 2018. Outstanding Challenges in the Transferability of Ecological Models. *Trends in Ecology & Evolution* 33, 790–802. <https://doi.org/10.1016/j.tree.2018.08.001>
- Zurell, D., Elith, J., Schröder, B., 2012. Predicting to new environments: tools for visualizing model behaviour and impacts on mapped distributions: Predicting to new environments. *Diversity and Distributions* 18, 628–634. <https://doi.org/10.1111/j.1472-4642.2012.00887.x>

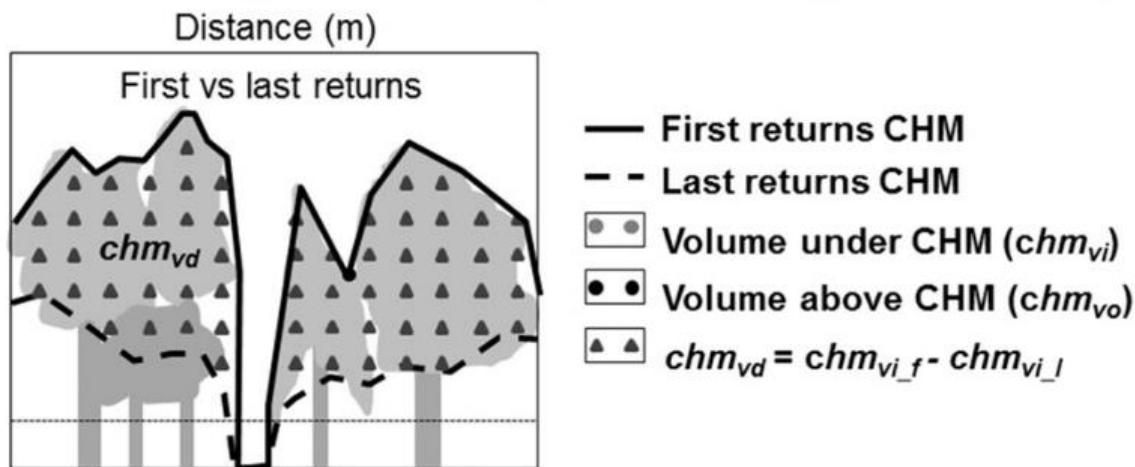
MERCI !

RESULTS: Local models

Forêt	Modèle retenu
Deodatie	$Ba \sim VolIn + Canopy_Closure + Zmean + B11 + Forest_type$
Deux Lacs	$Ba \sim VolIn + Zq5 + Zq10 + Zq20 + B11$
Donon	$Ba \sim VolIn + GapRatio + p2th + B11 + Forest_type$
Gueberschwihr	$Ba \sim VolIn + Zq30 + GapRatio + B11 + B12 + Forest_type$
Mouthierhouse	$Ba \sim VolIn + gapRatio + Zq50 + B11 + Forest_type$
Onf Observatory	$Ba \sim VolIn + GapRatio + B11 + Forest_Type$
NFI	$Ba \sim VolIn + Zmean + GapRatio + B11 + Forest_type$

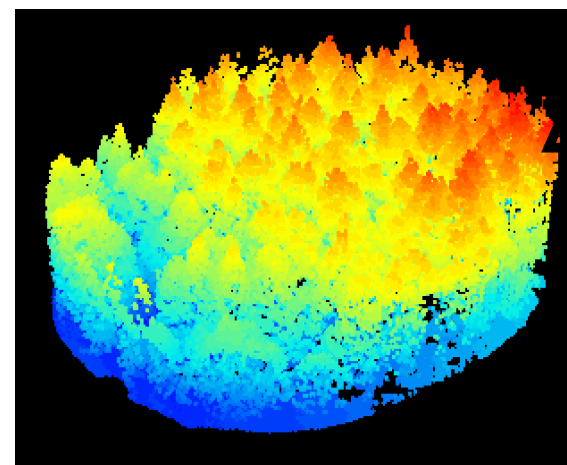
- ✓ **Sentinel Bands** 11 & 12 : Infrarouge à ondes courtes (SWIR)
- ✓ **Gap** Occupation within the plots
- ✓ **Forest Type**: 1: Broadleaves, 2: Mixed, 3: Conifers

✓ CANOPY HEIGHT MODEL

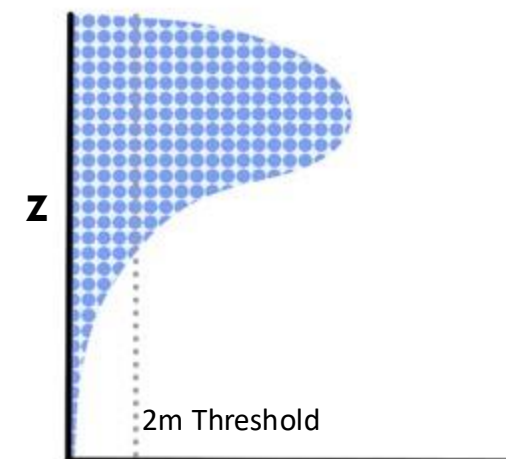


(Véga et al., 2016).

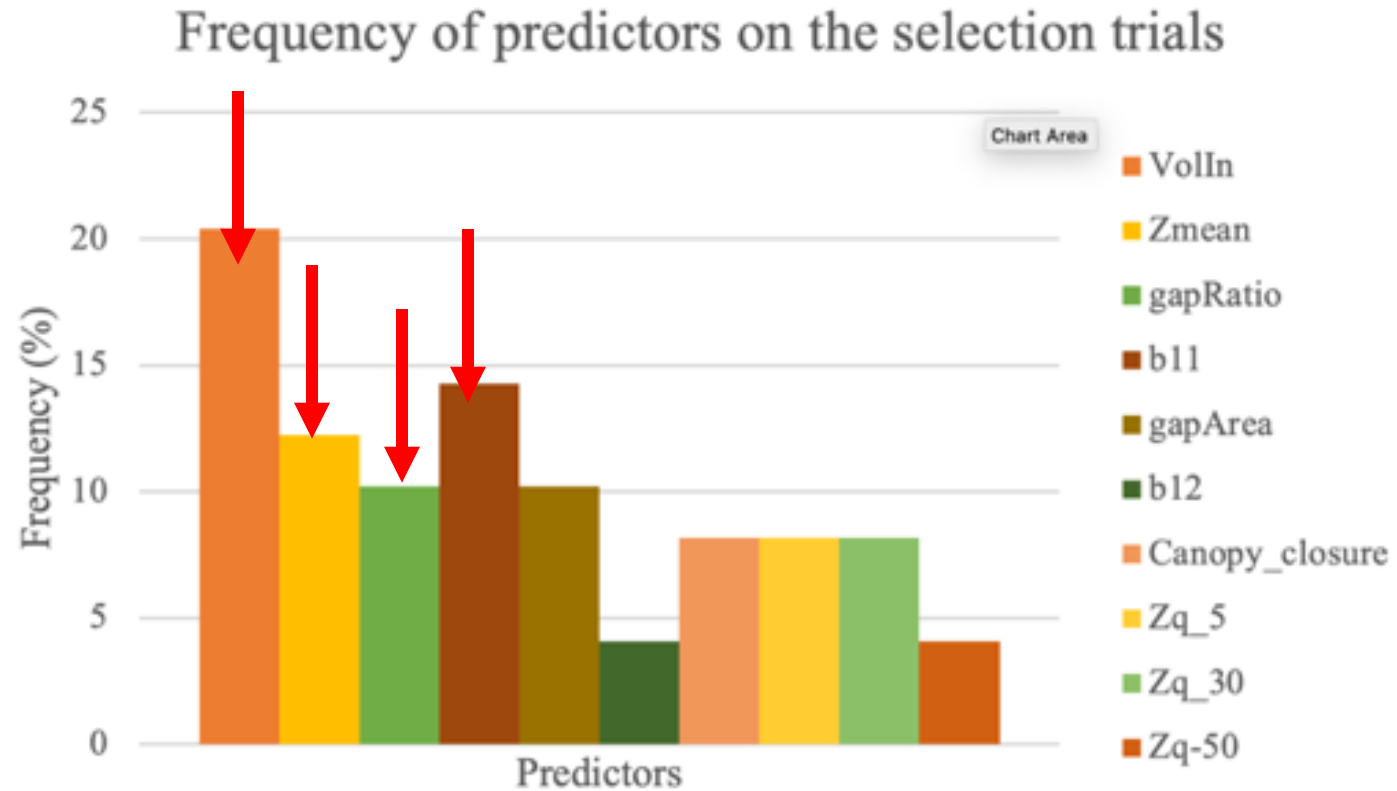
✓ ZMEAN: Hauteur moyenne (z)-nuage de points



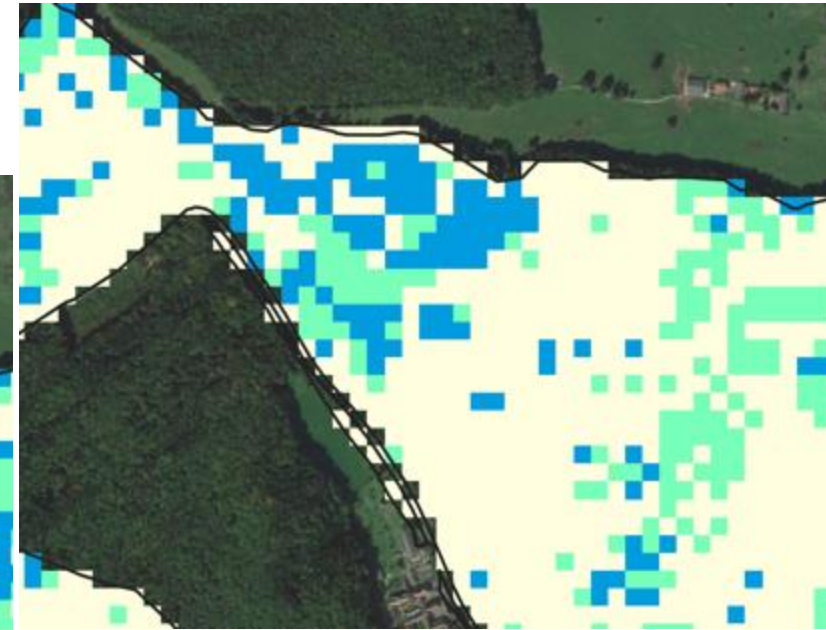
liDR Package Rstudio



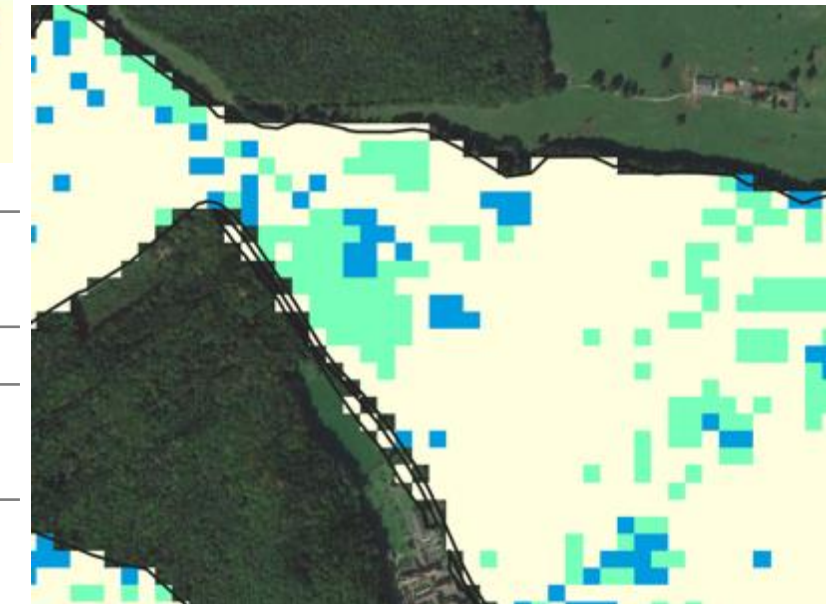
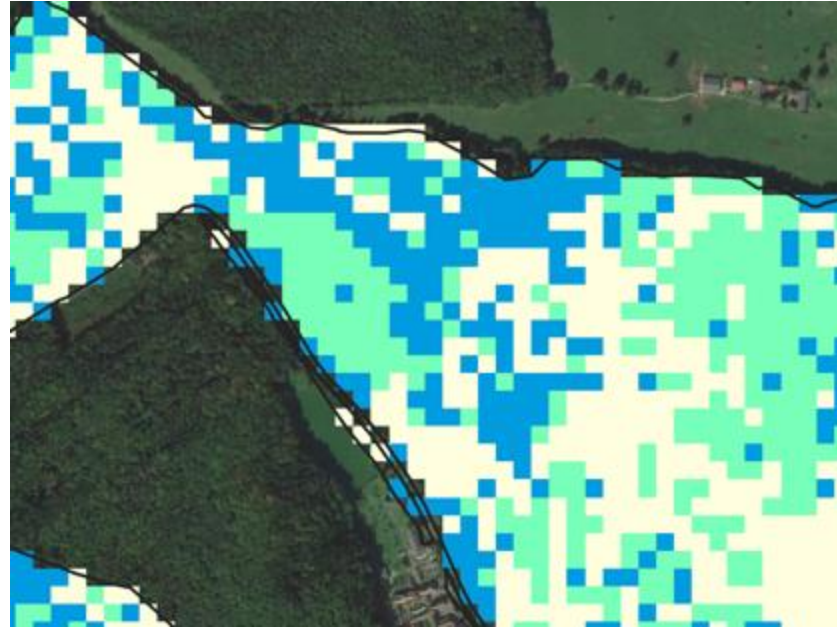
Annex 1. Models



ONF model



Gueberschwhir model



NFI model



Forest	R2	N° Plots in Ft 1	N° Plots in Ft 2	N° Plots in Ft 3
Gueberschwhir	79%	110	126	166
Observatory - > Gueberschwhir	43%	61	52	85
NFI - > Gueberschwhir	28%	181	145	259

RESULTS: EXTRAPOLATED PIXELS

Over the AOI

(Far from Convex Hull)

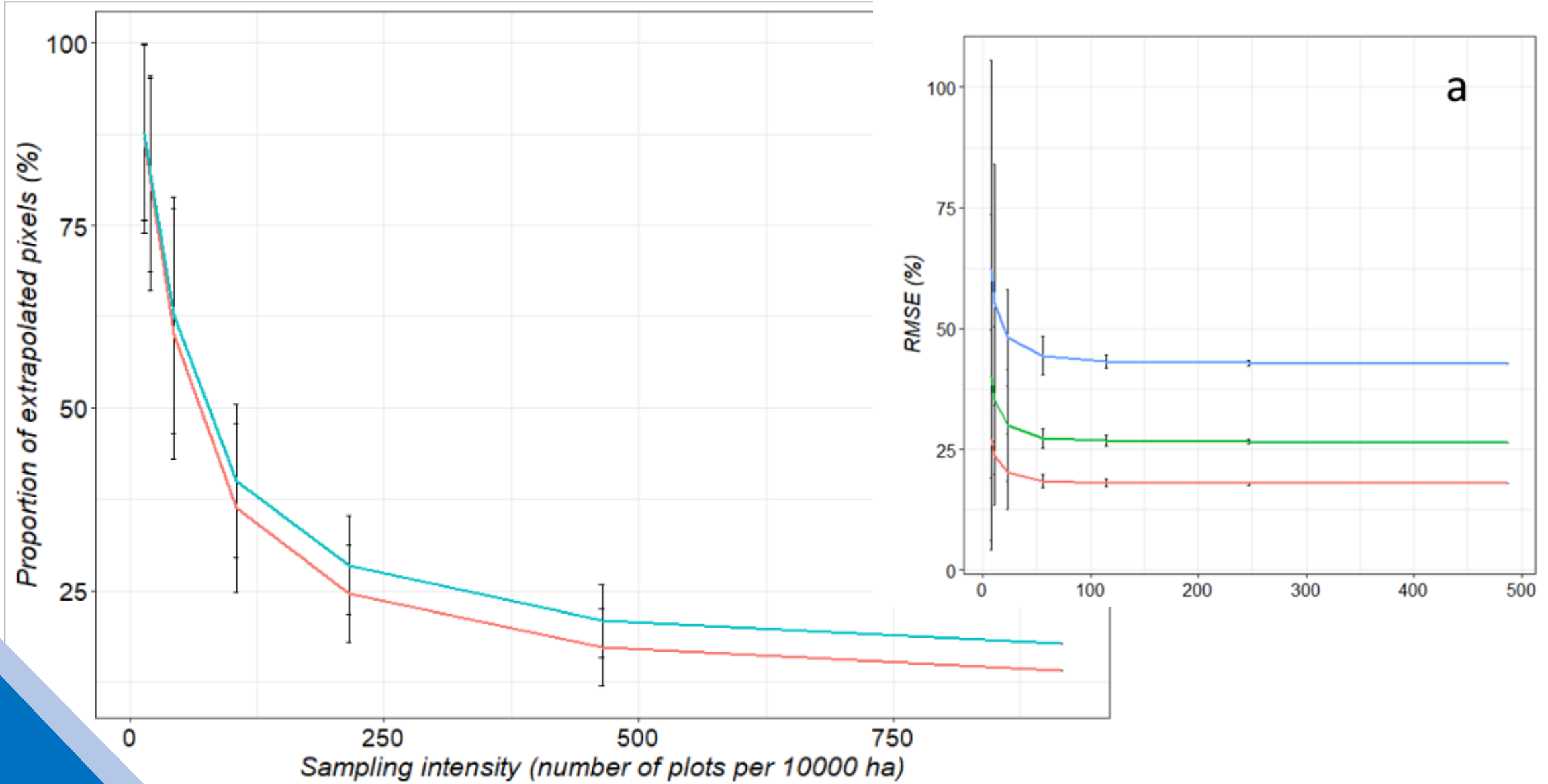
		MODELS						
FOREST	Nº Pixels	Deodatie	Deux lacs	Donon	Gueberschwihr	Mouterhouse	LSMFI	NFI
Deodatie	369 648	36,3 %	62,2 %	34,7 %	84,6 %	56,6 %	27,7 %	24,5 %
Deux lacs	6 519	36,8 %	36,7 %	36 %	95,2 %	68,9 %	23 %	25,5 %
Donon	126 480	45,8 %	62,3 %	37 %	85,3 %	53,5 %	29,1 %	26,8 %
Gueberschwihr	4 616	27,9 %	63,6 %	19,7 %	26,2 %	14,6 %	12,9 %	8,8 %
Mouterhouse	58 047	50,7 %	78,5 %	38 %	64 %	20,4 %	25,2 %	14,1 %
AOI	3 917 489	43,6 %	72,7 %	41,2 %	74,2 %	41,5 %	29,1 %	21,1 %
Mean		40,9 %	67,8 %	33,9 %	80,7 %	47,0 %	23,6 %	24,2 %

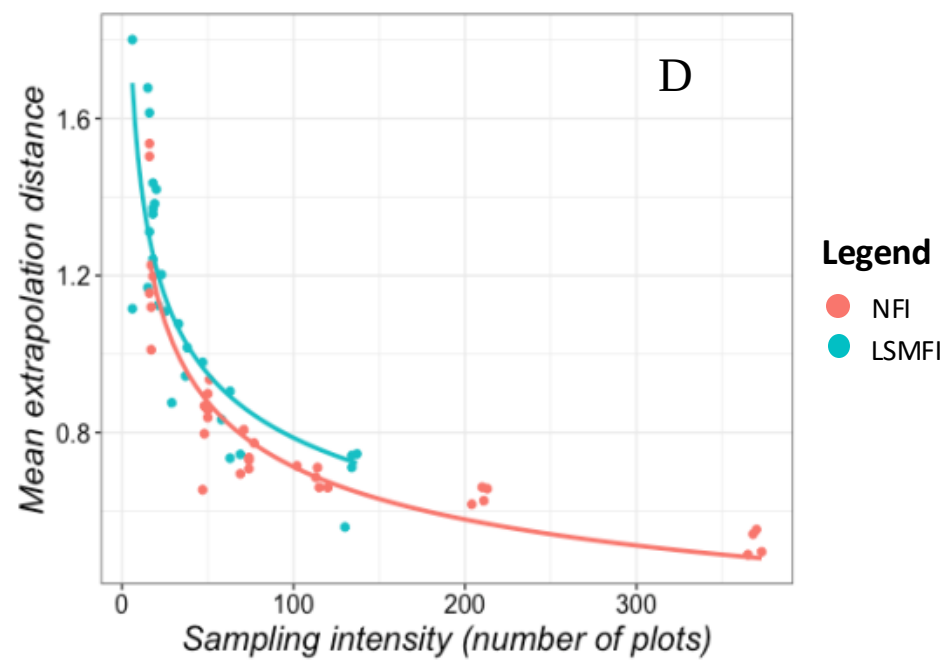
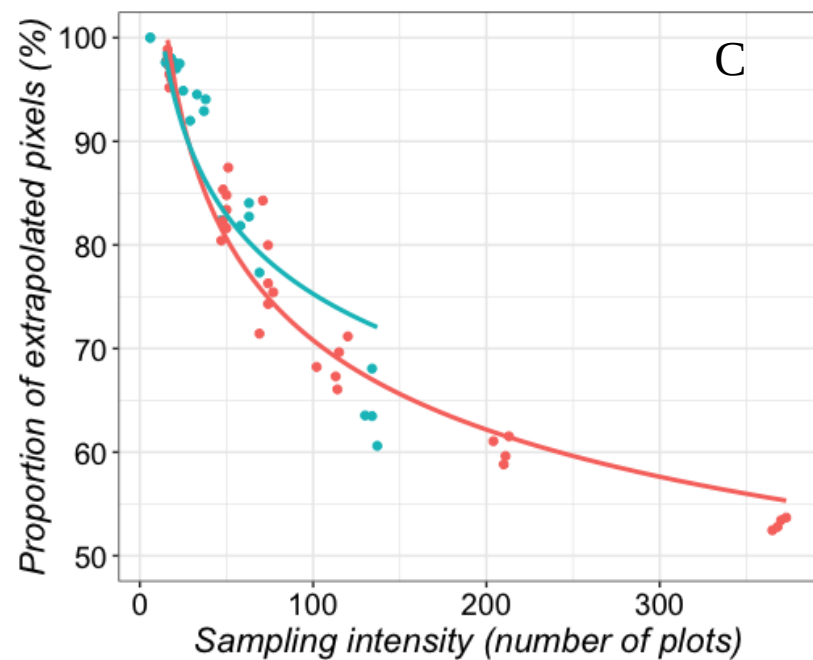
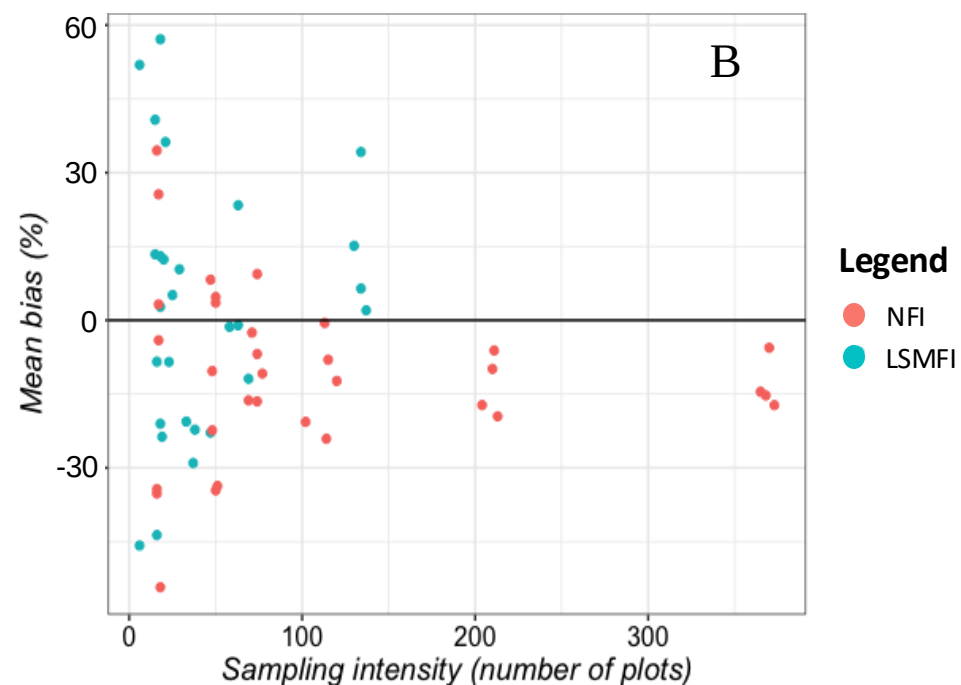
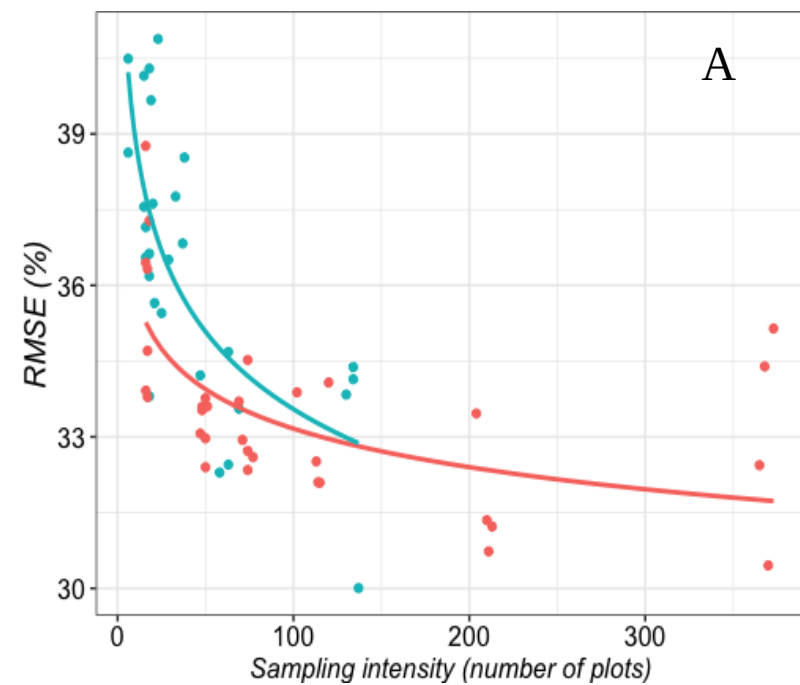
Plot details

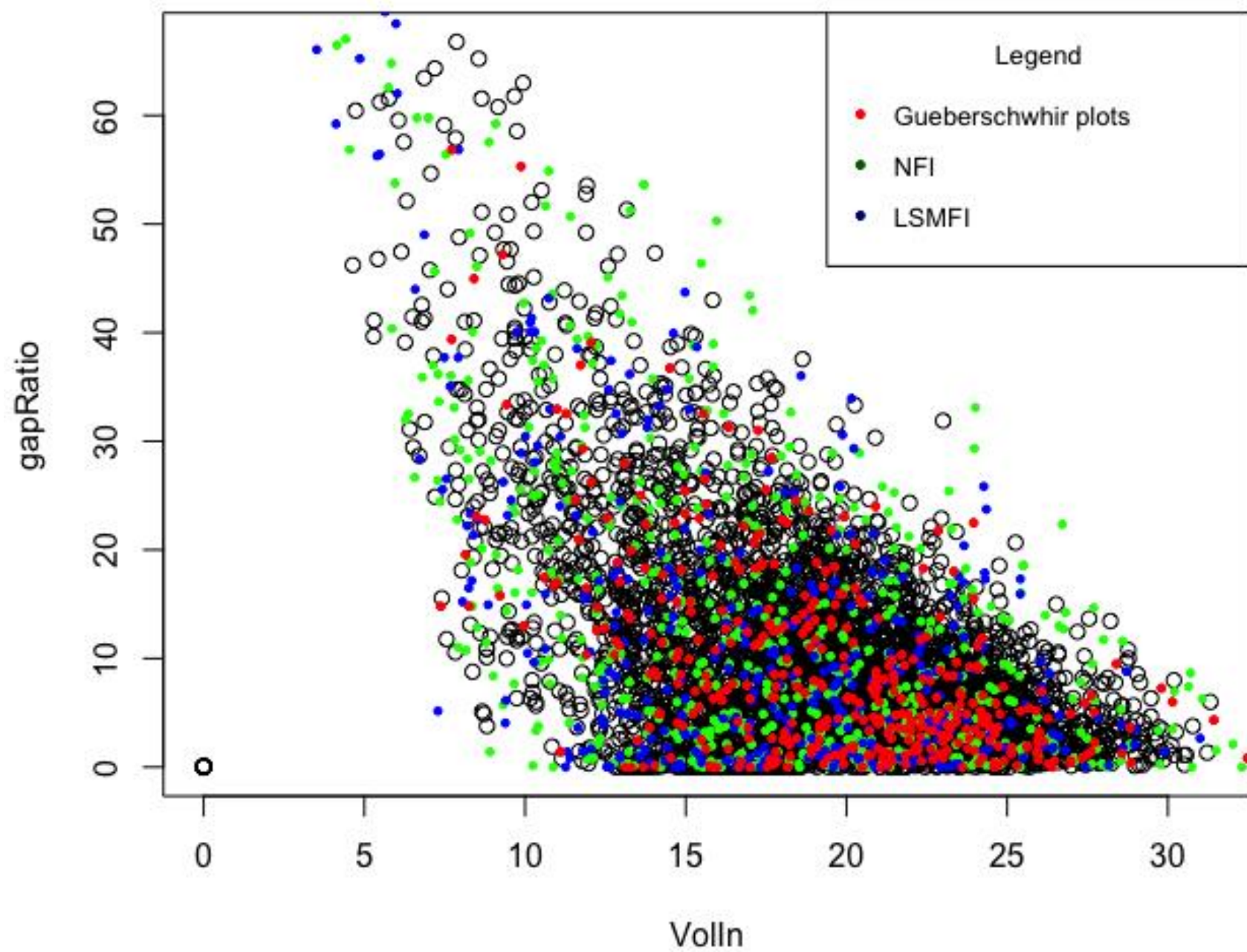
Forest	Year of the field survey	N° of plots	Mean Ba (m ² /ha)	Sd of Ba	N° Plots in Ft 1	N° Plots in Ft 2	N° Plots in Ft 3
Deodatie	2019	192	27.2	13.9	31	40	121
Deux lacs	2020	346	27.0	12.7	0	17	329
Donon	2020-2021	561	26.8	9.8	119	77	365
Gueberschwihr	2020-2021	402	28.2	8.6	110	126	166
Mouthouse	2019	544	23.6	8.3	280	89	175
Observatory	2021	198	25.6	11.9	61	52	85
NFI	2016-2020	585	30.6	13.8	181	145	259

FT1= Broadleaves / FT2: Mixed / FT3= Conifers

On Mouterhouse Forest







Quelle est la fiabilité des cartes de télédétection calibrées sur de vastes zones ?

.. avec un échantillon **insuffisant**, une représentation biaisée de la population peut souvent se produire
..(Bartley et al,2019)

...C'est important de **localiser** les prédictions spatiales incertaines afin d'accroître la **confiance** dans la cartographie produite à grande échelle ...(Meyer and Pebesma, 2021)

...Du point de vue de la gestion opérationnelle des forêts, le fait de savoir où les prédictions sont hors de leur **domaine de validité** pourrait représenter un résultat utile de cette approche... (Renaud et al, 2022)

